University of Florida, Department of Mathematics

Analysis First-Year Exam, August 2021, Part 2

Answer FOUR questions in detail. State carefully any results used without proof.

1. Let f_A be the indicator function of the countably infinite set $A \subseteq [0,1]$: that is, $f_A(t)$ is 1 when $t \in A$ and 0 when $t \in [0,1] \setminus A$.

(i) Exhibit (with brief justification) such an A for which f_A is Riemann integrable, or prove that no such A exists.

(ii) Exhibit (with brief justification) such an A for which f_A is not Riemann integrable, or prove that no such A exists.

2. For each $n > 0$ let the function $f_n : X \to \mathbb{R}$ be continuous at all but finitely many points. Prove that if $f_n \to f$ uniformly on X, then f is continuous at all but countably many points.

3. $(f_n : n > 0)$ is an equicontinuous sequence of real-valued functions on a compact space. Prove that if $f_n \to f$ pointwise then $f_n \to f$ uniformly.

4. Let $(f_n : n > 0)$ be a sequence of measurable functions on Ω. Prove that each of the following sets is measurable:

(i) $\{\omega \in \Omega : \text{ the sequence } f_n(\omega) \text{ is eventually constant}\}$;

(ii) $\{\omega \in \Omega : \text{ the values } f_n(\omega) \text{ are all different}\}$.

5. Let $(f_n : n > 0)$ be a sequence of non-negative integrable functions that converges pointwise to f. Prove that if

$$\int_\Omega f_n \, d\mu \to \int_\Omega f \, d\mu$$

then

$$\int_\Omega |f - f_n| \, d\mu \to 0.$$

[It might help to consider the positive part $(f - f_n)^+ = \max\{0, f - f_n\}$.]